Valve over lap

The portion of the operating cycle in which, when the piston is passing TDC (top dead centre) on the exhaust stroke, both the intake and exhaust valves are open. It is usually expressed in degrees of crankshaft rotation. Valve overlap is necessary for the efficient flow of gases in and out of the combustion chamber. It is will vary with respect to size and configuration of engine.

5.8 TWO STROKE DIESEL ENGINE

When the cycle of operation of engine completes in one revolution of the crankshaft or two stroke of the piston and one power stroke is obtained in each revolution of a crankshaft, then the engine is called as two stroke engine.

Construction

Inside the engine, the small end of the connecting rod is connected to the piston with the help of piston pin. The big end of the connecting rod is connected to the crankpin of the crankshaft. When the crankshaft rotates, the piston will reciprocate on the cylinder and vice versa. There is no inlet and exhaust valves as in case of four-stroke engine but consists of the inlet port (IP), an exhaust port (EP) and transfer port (TP). The fresh charge enters into the crankcase through inlet port. Transfer port is used to transfer the compressed charge from crankcase to the cylinder. Exhaust port is used to transfer the burnt gases out of the engine. The movement of the piston will open and closes the ports. A fuel injector is placed in the engine head.

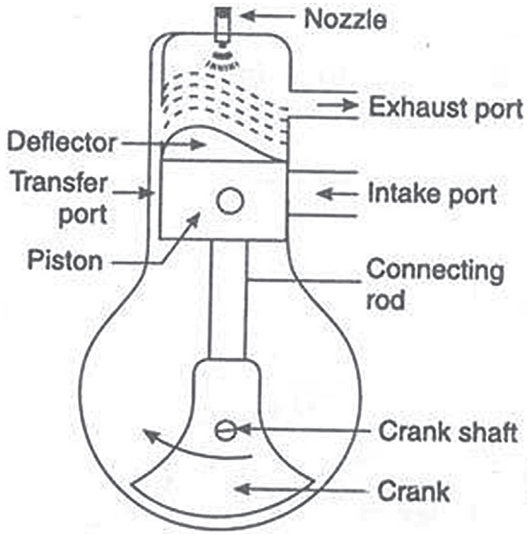


Figure 5.8(a) Two Stroke Diesel Engine

Working principle

In two stroke engine, all the four events namely suction, compression, power and exhaust cannot be distinctly identified. For each revolution of crankshaft or for every two stroke of piston, a working / power stroke is obtained. Hence the cycle of operation can be explained with two stroke of piston movement namely Upward stroke and Downward stroke.

Upward Stroke

During upward stroke, the piston moves from BDC to TDC. It expels the burnt gases to the atmosphere through the exhaust port. It closes the transfer port and then the exhaust port. Then it compresses the already inducted charge (air-fuel mixture) in the combustion chamber of the cylinder. At the end of the upward stroke, the diesel fuel is injected at high pressure through fuel injection nozzle and this will automise the fuel. The temperature of the compressed air is sufficient enough to ignite the diesel and combustion is take place.

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